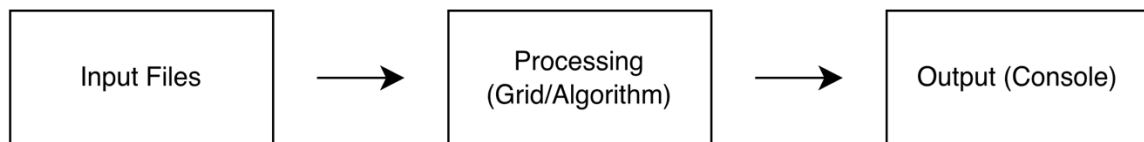


Weather Information Processing System

1. System Overview

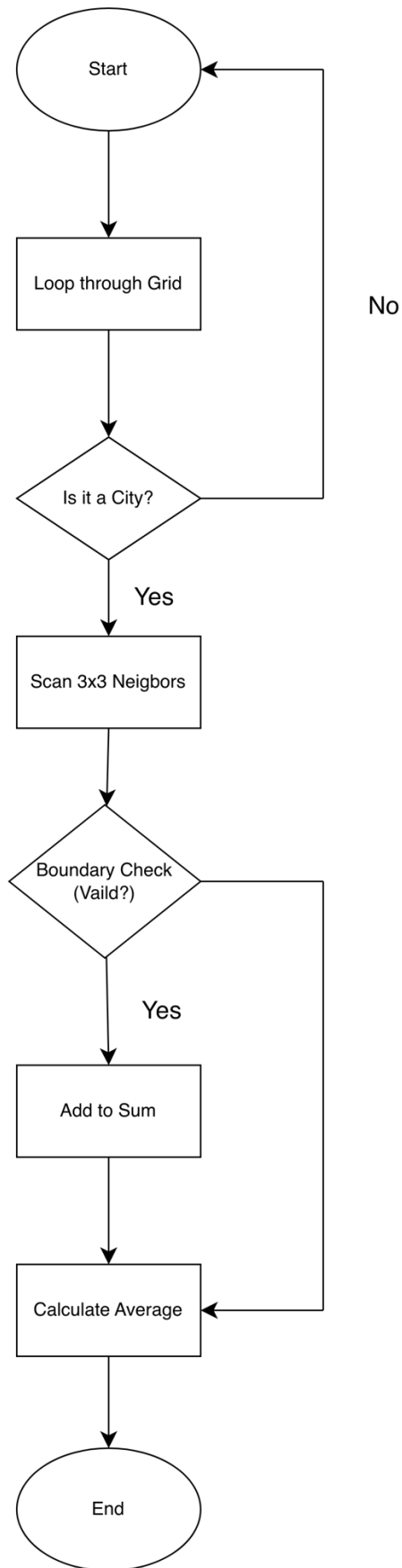
The Weather Information Processing System is designed based on a modular architecture. It consists of three main components.

1. Input Module (FileReader): Reads the configuration file (config.txt) and parses specific weather data files (city location, cloud cover, pressure) into memory,
2. Processing Module (WeatherForecast): Manages the core data structure (GridCell**) and executes the weather prediction algorithm using a 3x3 spatial scan.
3. Output Module (Display): Renders visual maps and generates the final weather forecast summary report to the console.



2. Algorithm Flowchart (generateForecasts)

The following flowchart illustrates the logic of the generateForecasts function. It iterates through the entire grid. Upon finding a city, it triggers a nested loop to scan the surrounding 9 cells. A boundary check is performed to ensure array indices remain valid before calculating averages.



3. Test Execution Screenshots

```
==8892== Memcheck, a memory error detector
==8892== Copyright (C) 2002-2017, and GNU GPL'd, by Julian Seward et al.
==8892== Using Valgrind-3.18.1 and LibVEX; rerun with -h for copyright info
==8892== Command: ./csci251_a1.app
==8892==
Please enter your Student ID: 9890464
Please enter your Student Name: Jin Wan Kim

Student ID   : 9890464
Student Name : Jin Wan Kim
-----
Welcome to Weather Information Processing System!

1) Read in and process a configuration file
2) Display city map
3) Display cloud coverage map (cloudiness index)
4) Display cloud coverage map (LMH symbols)
5) Display atmospheric pressure map (pressure index)
6) Display atmospheric pressure map (LMH symbols)
7) Show weather forecast summary report
8) Quit

Please enter your choice : 1
[ Read in and process a configuration file ]
Please enter config filename : config.txt
Reading in GridX_IdxRange ... done!
Reading in GridY_IdxRange ... done!

Storing data from input file :

All records successfully stored.

Press <enter> to go back to main menu ...

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Please enter your choice : 1
[ Read in and process a configuration file ]
Please enter config filename : wrong.txt
[Error: Cannot open wrong.txt
Error reading config!

Press <enter> to go back to main menu ...
```

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Please enter your choice : 1

[Read in and process a configuration file]

Please enter config filename : test_config.txt

Reading in GridX_IdxRange ... done!

Reading in GridY_IdxRange ... done!

Storing data from input file :

All records successfully stored.

Press <enter> to go back to main menu ...

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Please enter your choice : 2

```
# # # # # # # # # # #  
8 #      2 2      #  
7 #      2 2      1 #  
6 #                      #  
5 #                      #  
4 #                      #  
3 #      3 3 3      #  
2 #      3 3 3      #  
1 #      3 3 3      #  
0 #                      #  
  # # # # # # # # # #  
  0 1 2 3 4 5 6 7 8
```

Press <enter> to go back to main menu ...

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Please enter your choice : 3

```
# # # # # # # # # # # #  
8 # 8 2 3 2 7 7 3 8 1 #  
7 # 3 7 3 5 2 9 1 3 0 #  
6 # 4 8 7 5 4 2 1 1 4 #  
5 # 3 3 4 2 7 4 4 6 1 #  
4 # 7 5 5 6 4 1 2 6 3 #  
3 # 2 1 7 9 8 5 9 1 2 #  
2 # 9 4 3 2 4 3 3 0 6 #  
1 # 9 6 8 1 6 4 6 0 2 #  
0 # 4 3 5 8 7 1 6 4 0 #  
# # # # # # # # # #  
0 1 2 3 4 5 6 7 8
```

Press <enter> to go back to main menu ...

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- 8) Quit

Please enter your choice : 4

```
# # # # # # # # # # #  
8 # H L L L H H M H L #  
7 # M H M M L H L M L #  
6 # M H H M M L L L M #  
5 # M M M L H M M M L #  
4 # H M M H M L L M M #  
3 # L L H H H M H L L #  
2 # H M L L M M L L M #  
1 # H H H L M M M L L #  
0 # M M M H H L H M L #  
# # # # # # # # # # #  
  0 1 2 3 4 5 6 7 8
```

Press <enter> to go back to main menu ...

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- 8) Quit

Please enter your choice : 5

```
# # # # # # # # # # # #  
8 # 5 8 3 3 2 2 8 3 7 #  
7 # 4 3 4 3 6 5 9 2 8 #  
6 # 5 0 1 1 6 8 0 8 4 #  
5 # 6 8 1 5 7 4 5 5 7 #  
4 # 0 8 7 1 4 6 6 7 2 #  
3 # 6 2 1 1 8 4 4 7 1 #  
2 # 3 2 1 3 5 3 4 6 7 #  
1 # 3 6 4 8 5 3 5 5 3 #  
0 # 3 2 7 5 1 5 2 3 7 #  
# # # # # # # # # # #  
0 1 2 3 4 5 6 7 8
```

Press <enter> to go back to main menu ...

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- 8) Quit

Please enter your choice : 6

```
# # # # # # # # # # #
8 # M H M M L L H L H #
7 # M M M M H M H L H #
6 # M L L L H H L H M #
5 # M H L M H M M M H #
4 # L H H L M H M H L #
3 # M L L L H M M H L #
2 # M L L L M M M M H #
1 # L H M H M M M M L #
0 # M L H M L M L M H #
  # # # # # # # # # #
    0 1 2 3 4 5 6 7 8
```

Press <enter> to go back to main menu ...

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- 8) Quit

Please enter your choice : 7

[Weather Forecast Summary Report

City Name : Big_City
City ID : 3
Ave. Cloud Cover (ACC) : 60.44 (M)
Ave. Pressure (AP) : 39.22 (M)
Probability of Rain (%) : 50.00

~~~~~  
~~~~~  
\\
~~~~~  
~~~~~  
\\

City Name : Mid_City
City ID : 2
Ave. Cloud Cover (ACC) : 50.89 (M)
Ave. Pressure (AP) : 35.00 (M)
Probability of Rain (%) : 50.00

~~~~~  
~~~~~  
\\
~~~~~  
~~~~~  
\\

City Name : Small_City
City ID : 1
Ave. Cloud Cover (ACC) : 29.56 (L)
Ave. Pressure (AP) : 57.67 (M)
Probability of Rain (%) : 40.00

~~~~~  
~~~~~  
\\\
~~~~~  
~~~~~  
\\

Press <enter> to go back to main menu ...

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Please enter your choice : 8

Exiting program...

Memory successfully deallocated!

==8892==

==8892== HEAP SUMMARY:

==8892== in use at exit: 0 bytes in 0 blocks

==8892== total heap usage: 74 allocs, 74 frees, 117,724 bytes allocated

==8892==

==8892== All heap blocks were freed -- no leaks are possible

==8892==

==8892== For lists of detected and suppressed errors, rerun with: -s

==8892== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)